Pratheek Nistala

Chennai, India
+91-9391161436 | prtk24.work@gmail.com | linkedin.com/in/xyzprtk | github.com/xyzprtk | prtx.zyz

Professional Summary

Data Scientist with 11 months of experience working on production credit risk models at Dun & Bradstreet. Built and evaluated XGBoost-based segmentation models and engineered features on 5M+ record datasets. Skilled in Python, SQL, and PySpark, with project experience across fintech and e-commerce domains.

Technical Skills

Programming Languages: Python, SQL, R, Java
Machine Learning & Modelling: Scikit-learn, XGBoost, TensorFlow, Random Forest, Logistic Regression
Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Microsoft Excel
Big Data & Distributed Computing: PySpark, Apache Spark, Databricks
Development Tools: Git, Jupyter Notebook, Bash Scripting
Operating Systems: Linux, Windows
Core Competencies: Data Manipulation, Analytical Ability

Education

Vellore Institute of Technology — **Sep. 2021 – May 2025**
Bachelor of Science in Computer Science with Data Science — *Vellore, India*
CGPA: 8.3/10
Relevant Coursework : Machine Learning, Statistical Modeling, Exploratory Data Analysis, Database Management Systems, Big Data Analytics, Software Engineering

Experience

Dun & Bradstreet — **June 2025 – Present**
Data Science Apprentice (Data Analyst / Risk Analytics) — *Chennai, India*

- Collaborated with the data science team on **Delinquency Score Model** by leading **feature engineering** efforts and applying analytical ability to enable an **XGBoost**-driven probability-to-score mapping (101 to 670) with Class 1 to 5 risk segmentation, improving model discriminatory power (**AUC** +4, **KS** +7).
- Developed **scorecard-style models** to convert predictive outputs into business-interpretable risk scores.
- Conducted **SQL-driven EDA** and **data quality validation** on large datasets (5M+ records) to support modeling workflows.
- Performed **performance validation** of external vendor data against internal risk models.

Projects

Credit Risk Scorecard Platform | *Next.js, FastAPI, Python, Scikit-learn, XGBoost* — **GitHub** | **Live** | **Blog**

- Built an end-to-end credit risk scorecard platform on **149K+ borrower records**, achieving **0.859 AUC** and **0.567 KS statistic** using WoE-based Logistic Regression.
- Designed a preprocessing and feature engineering pipeline with **optimal binning, outlier capping, and missing-value handling**, improving score distribution consistency across risk tiers.
- Developed a full-stack application using **Next.js and FastAPI** with real-time scoring APIs, feature-level score breakdowns, and responsive dashboards for interactive risk analysis.

Tech Mental Health Analysis | *Python, Pandas, NumPy, Scikit-learn, Seaborn* — **GitHub** | **Blog**

- Analyzed mental health survey data from **10,000+ technology professionals** to identify key behavioral and workplace-related trends.
- Performed **EDA and feature engineering** to extract predictors influencing treatment-seeking behavior.
- Built and optimized **machine learning models** (Random Forest, Logistic Regression, Decision Tree), achieving **80%+ prediction accuracy**.

e-Commerce Customer Segmentation & Prediction | *Python, Pandas, NumPy, Scikit-learn, Seaborn* — **GitHub** | **Blog**

- Analyzed **100,000+ e-commerce orders** (2016–2018) to uncover trends in customer behavior, geolocation impact, and order performance.
- Applied **RFM analysis and K-Means clustering** to segment customers into high-, medium-, and low-value cohorts.
- Predicted **customer review scores with 80%+ accuracy** using machine learning models based on order cost, delivery performance, and location features.